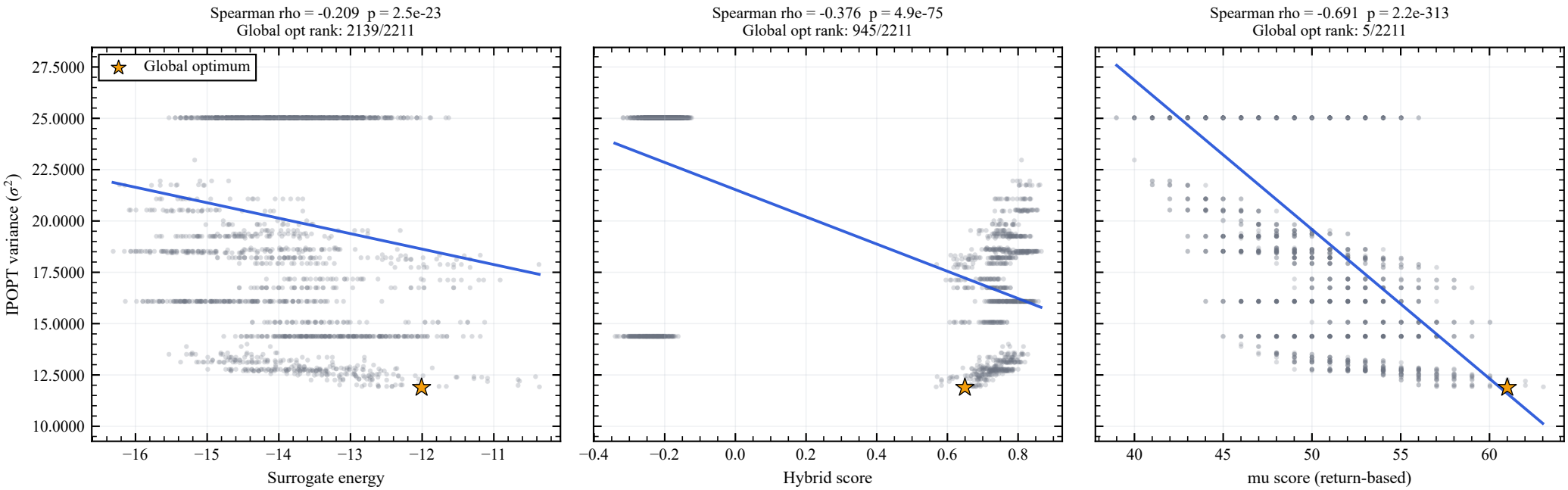

Surrogate Alignment with IPOPT-Refined Portfolio Variance
Spearman rho = -0.209 p = 2.5e-23
Global opt rank: 2139/2211
Spearman rho = -0.376 p = 4.9e-75
Global opt rank: 945/2211
Spearman rho = -0.691 p = 2.2e-313
Global opt rank: 5/2211
Global optimum
IPOPT variance (σ^2)
27.5000
25.0000
22.5000
20.0000
17.5000
15.0000
12.5000
10.0000
−16
−15
−14
−13
−12
−11
Surrogate energy
−0.4
−0.2
0.0
0.2
0.4
0.6
0.8
Hybrid score
40
45
50
55
60
mu score (return-based)